MILITARY SPECIFICATION

GUN, AIR DEFENSE ARTILLERY 20MM: TOWED M167

1. SCOPE

1.1 This specification covers one type of antiaircraft weapon system which consists of an electrically powered six barrel 20 millimeter (mm) automatic gun with linked ammunition and feed system, radar range and lead computing sight, mounted in a towed two wheel vehicle. The M167, 20mm Towed Antiaircraft Artillery Gun is hereafter referred to as the weapon system.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or requiest for proposal, form a part of this specificatinn to the extent specified herein.

SPECIFICATIONS

Military

MIL-W-13855	-	Weapons, Small Arms and Aircraft Armament Subsystems, General Specification For.
MIL-P-14232	-	Parts, Equipment and Tools for Army Material, Packaging and Packing of.
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of.
MIL-G-45937	-	Gun, 20MM, Automatic: M168.
MIL-R-50651	-	Range Radar Set AN/VPS-2.
MIL-S-60661	-	Sight, Lead-Computing, M61 for Gun, Antiaricraft Artillery, Self-Propelled, 20MM, M163 and Towed, 20MM, M167.
MIL-G-60662	-	Generator, Sight Current.
MIL-F-60664	-	Fire Control for Gun, Antiaircraft Artillery, Towed, 20MM, M167 and Gun, Antiaircraft Artillery, Self-Propelled 20MM, M163.

FSC 1005

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STANDARDS

Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	-	Quality Assurance Terms and Definitions.
MIL-STD-461	-	Electromagnetic Interference Characteristics, Requirements for Equipment.
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurements of.
MIL-STD-810	-	Environmental Test Methods.

DRAWINGS

U.S. Army Weapons Command

F8438300	-	Gun, Antiaircraft Artillery, Towed: 20MM, M167.

PUBLICATIONS

U.S. Army Weapons Command

P8430372	-	Packaging Data Sheet for Gun, Antiaircraft Artillery, Towed: 20MM, M167.
IEL8430372	-	Index of Inspection Equipment.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. Requirements for submission of first article shall be as specified in the contract (see 6.2). Unless otherwise specified (see 6.2), the first article shall be included in the pilot pack (see 5.1).

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3.2 Material, construction and design. The weapon system shall conform to the material, construction and design requirement specified herein, on Drawing F8438300 and MIL-W-13855.

3.2.1 Gun, M168. The M168 20mm Automatic Gun shall be in accordance with MIL-G-45937.

3.2.2 Lead computing sight M61. The M61 lead computing sight shall be in accordance with MIL-S-60661 and MIL-F-60664.

3.2.3 Sight current generator. The sight current generator shall be in accordance with MIL-G-60662 and MIL-F-60664.

3.2.4 Range radar AN/VPS-2. The range radar shall be in accordance with MIL-R-50651 and MIL-F-60664.

3.2.5 Running gear. The weapon system running gear wheel nuts shall be torqued in accordance with the applicable drawing requirements and tire pressure shall be in accordance with drawing requirements.

3.2.6 Rear lights. The rear lights, electrical harness, and connector shall be installed in accordance with drawing requirements. There shall be no shorts, and continuity shall be in accordance with drawing requirements.

3.2.7 Hydraulic system. The hydraulic system shall show no evidence of oil leakage at the hydraulic fittings and connections when under operating pressure in accordance with the drawing requirements.

3.3 Performance characteristics.

3.3.1 Warm-up. The weapons system shall require no more than four minutes at +60°F or above, and one hour at -40°F to warm up to become operable in the test, radar, manual, external or ground mode. Testing shall be as specified in 4.5.3.2.

3.3.1.1 Test mode. The weapons system when placed in the TEST MODE shall indicate that the weapon system is operational by the illumination of the test completed light. Testing shall be as specified in 4.5.3.2.

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3.3.1.2 Radar mode. The weapon system when placed in the RADAR MODE shall permit the following functions when actuating either or both of the action switches located in the hand grips: Testing shall be as specified in 4.5.3.2.

a. Slew the gun and gun mount to acquire the target in the reticle of the lead-computing sight.
b. Cause the radar to radiate by depressing and holding the radar-radiate foot switch while continuing to track the target. The ready-to-fire indicator shall be energized when the target is acquired.
c. Fire the gun by depressing the right-hand trigger switch.
d. Return the system to the standby condition by releasing the trigger switch, the action switches, and the radar-radiate foot switch.

3.3.1.3 Manual mode. The weapon system when placed in the MANUAL MODE shall permit the following functions. Testing shall be as specified in 4.5.3.2.

a. The sight reticle shall have correct directional response to each potentiometer setting.
b. Slew the gun and gun mount to acquire the target in the sight reticle after actuating either or both of the action switches located in the hand grips.
c. Complete the firing circuit by depressing the trigger switch and actuating either or both of the action switches located in the hand grips.

3.3.1.4 External mode. The weapon system when placed in the EXTERNAL MODE and the remote control assembly installed. The remote operator shall provide the range setting, and energize the ready-to-fire indicator lamp in the sight. Corresponding settings in the external mode shall result in the same directional response obtained in the manual mode. Testing shall be as specified in 4.5.3.2.

3.3.1.5 Ground mode. The weapon system when placed in the GROUND MODE shall permit the following functions. Testing shall be as specified in 4.5.3.2.

a. The sight to be mechanically caged at a superelevation of -7 ± 2 milliradians.
b. Illumination of the sight reticle.

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c. An operator to slew, elevate, and aim the gun after actuating either or both of the action switches located in the hand grips.
d. An operator to fire the gun by depressing the trigger switch while actuating either or both of the action switches located in the hand grip.

3.3.2 System reaction time. With the following two combat conditions present, the weapon system shall slew 180 degrees and acquire, track and fire on the target in not more than 10.5 seconds. Testing shall be as specified in 4.5.3.2.

a. Target velocity and lead angle equivalent to a sight sensitivity, as defined in MIL-S-60661, of less than three seconds. The target shall have the following characteristics:

Speed	350 knots
Altitude	500 ± 50 meters
Crossover ground range	100 to 500 meters

b. Target within range and of a sufficient size for radar lock-on.

3.3.3 System fire control. The weapon system fire control operation shall be in accordance with MIL-F-60664 and as specified in the contract (see 6.2). Testing shall be as specified in 4.5.3.2.

3.3.4 Rates of fire. The weapon system shall fire the following two rates at +70°F + 30° ambient when selected by rotating the FIRING RATE switch. Testing shall be as specified in 4.5.3.2.

a. Low rate of fire 1000 ± 300 shots per minute (spm).
b. High rate of fire 3000 to 4000 spm.

3.3.5 Acceleration to firing rate. The weapon system shall attain a rate of 2700 spm from initial gun rotation within 0.35 seconds. Testing shall be as specified in 4.5.3.2.

3.3.6 Burst limit. The weapon system in the high rate of fire condition, using burst limiting switch, shall be capable of selecting four separate burst limits. The minimum and maximum rounds to be fired are as follows: Testing shall be as specified in 4.5.3.2.

a. Hi-Burst Limit 10: 8 to 12 rounds
b. Hi-Burst Limit 30: 27 to 33 rounds
c. Hi-Burst Limit 60: 54 to 66 rounds
d. Hi-Burst Limit 100: 90 to 110 rounds

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3.3.7 Field of fire. The weapon system shall have a field of fire defined for both azimuth and elevation as follows: Testing shall be as specified in 4.5.3.2.

a. Azimuth: 360 degrees in either direction.
b. Elevation: from 5 to 7 degrees below to 80 to 82 degrees above the horizontal reference line of the carriage. The electrical switch assembly in the elevation drive system shall limit powered elevation (and depression) travel from one to three degrees before the mechanical stops contact.

3.3.8 Slewing rate. The weapon system slewing rate shall be at least 75 degrees per second in azimuth in the ground mode. The minimum slewing rate in all other modes shall be 60 degrees per second in azimuth. The slewing and tracking rate shall be 45 degrees per second in elevation. The azimuth velocity shall be measured during the fourth revolution of the gun mount with the gun mount making six revolutions total. Testing shall be as specified in 4.5.3.2.

3.3.9 Gun mount acceleration. With maximum hand control signal, the initial gun mount acceleration shall be at least 160 degrees per second squared in azimuth and 160 degrees per second squared in elevation when measured from the start of the gun mount movement. Initial acceleration is defined as the slope of a line through a point tangent to the velocity curve normalized at the first 0.2 second increment. Testing shall be as specified in 4.5.3.2.

3.3.10 Stall and stop. The weapon and system shall incur no damage when operated under the following conditions. Testing shall be as specified in 4.5.3.2.

a. When the gun mount and gun are in positive travel lock position.
b. When the gun is run into the elevation stops at the slewing rate specified in 3.3.8.

3.3.11 Firing interrupter. The firing interrupter shall interrupt firing in predetermined zones of azimuth and elevation when activated by extending the applicable rods in the firing interrupter assembly. Testing shall be as specified in 4.5.3.2.

3.3.12 Azimuth indicator. The azimuth indicator shall be free to be positioned manually and shall retain its manually preset position with respect to the turret during the firing of the weapon system. The lower pointer of the azimuth indicator shall be in zero degrees position when the gun barrels are in travel lock position. The upper two pointers shall move to correspond with the traverse of the gun mount and are resettable. Also

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the outer most ring called the "gunner aid dial" is resettable. Testing shall be as specified in 4.5.3.2.

3.3.13 Operational capability. The weapon system shall operate at both the high and low firing rates specified in 3.3.4 and acceleration to firing rates specified in 3.3.5 at any ambient temperature from -40°F to +125°F. The permissible degradation of the performance specified in 3.3.4 and 3.3.5 shall be 5, 10, and 15 percent respectively at 0°F, -20°F and -40°F. Testing shall be as specified in 4.5.3.2.

3.3.14 Declutching feeder. The declutching feeder shall delink the linked ammunition coming from the feed chute, feed the rounds into the gun, and guide the empty links into the exit chute. With the GUN CLEAR switch set in AUTO position, the declutching feeder shall completely disengage the ammunition feed system during the clearing cycle of the end of each burst cycle. Testing shall be as specified in 4.5.3.2.

3.3.15 Clearing.

3.3.15.1 Gun clearing switch (Power). The GUN CLEAR switch when set in the AUTO position, the weapon system shall operate automatically to clear the gun. The GUN CLEAR switch when held in the OPERATOR position, the gunner shall be capable of manually clearing the gun by depressing the action switch in the elevation handle. Testing shall be as specified in 4.5.3.2.

3.3.15.2 Brake-clear and brake switch (Manual). The BRAKE-CLEAR and BRAKE switch is a three position switch, which is springloaded to the OFF (center) position. The BRAKE switch when held in BRAKE position shall release the gun drive motor brake and allow rounds to be cycled through the gun by manually rotating the barrel cluster. The BRAKE-CLEAR switch when held in the BRAKE-CLEAR position shall release the gun drive motor, energize the declutching feeder solenoid which shall prevent rounds from being fed into the gun. Testing shall be as specified in 4.5.3.2.

3.3.16 Sustained use. The weapon system when operated with battery power only and the fire switch in the high position shall fire three 60 round bursts at seven minute intervals. The ambient temperature at which these firings are conducted shall be within 10°F of the ambient temperature at which the rates of fire were established in 3.3.4. The system battery power shall remain ON during the time required to perform the three firings. Degradation of the third burst shall not exceed 15 percent of the performance as recorded for the system in 3.3.4, 3.3.5 and 3.3.6 c. Testing shall be as specified in 4.5.3.2.

3.3.17 Boresighting. The weapon system shall be boresighted utilizing a distant aiming point of at least 2,500 meters. Testing shall be as specified in 4.5.3.2.

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3.3.18 Hand brakes. The weapon system hand brakes, when engaged, shall prevent the fully equipped and serviced weapon system from rolling on a 60 percent slope of brushed concrete. The brake shoes shall not drag when the brake lever is in the OFF position. Testing shall be as specified in 4.5.3.2.

3.4 Towing. The weapon system, prepared for travel, shall withstand four 1000 mile test cycles under conditions specified in Table I without damage or loosening of parts. The weapon system shall be capable of ascending and descending slopes of 60 percent and negotiating side slopes of 20 percent. The weapon system shall be capable of traversing ditches where the sum of the slopes forming the ditch does not exceed 60 percent and the straight away of each slope is at least as long as the combined length of the weapon system and mover. Testing shall be as specified in 4.4.

TABLE I

Road Type	Speed (MPH)	Distance (Miles)
Gravel	20 - 30	300
Belgian Block	15 - 20	50
Hilly Cross-country	5 - 15	200
Level Cross-country	10 - 20	250
Paved	45 Maximum	300
		Total 1000

3.4.1 Fording function. The weapon system shall be capable of fording a depth of 30 inches with leakage into the carriage not to exceed 473 cubic centimeters (one pint) per hour. All components and assemblies below the 30 inch fording depth, shall not suffer degradation in performance from water. Testing shall be as specified in 4.4.

3.5 Environmental. The weapon system shall meet the performance requirements of this specification as specified in 3.3 during or after exposure to environmental conditions specified in 3.5.1 through 3.5.8. Testing shall be as specified in 4.4.

3.5.1 Altitude. The weapon system shall operate without failure at altitudes from sea level to 10,000 feet and shall be transportable at altitudes up to 40,000 feet.

3.5.2 Temperature. The weapon system shall operate without failure in temperatures ranging from -40°F to +125°F. The weapon system shall function within the requirements of this specification after storage in a temperature of +155°F for a minimum of four hours.

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3.5.3 Rain. The weapon system shall show no degradation of performance or evidence of corrosion which would prevent the weapon system from performing its intended function with no protective cover and exposed to rain and wind conditions specified in Procedure I, Method 506 of MIL-STD-810.

3.5.4 Humidity. The weapon system shall show no degradation of performance when exposed to relative humidity of 5 percent at +125 F to a relative humidity of 100 percent at +85°F. The weapon system shall withstand storage in a relative humidity of 100 percent at +85°F for a minimum of 48 hours.

3.5.5 Fungus. The weapon system shall be resistant to microorganisms specified in Method 508 of MIL-STD-810.

3.5.6 Salt fog. The weapon system shall show no degradation of performance or evidence of corrosion which would prevent the weapon system from performing its intended function when exposed to the effects of a salt atmosphere as defined herein.

a. Temperature, +95°F.
b. Atomization of approximately 3 quarts of salt solution per 10 cubic feet of test chamber volume per 24 hours.
c. The weapon system shall be exposed to the salt fog atmosphere for a period of 48 hours.

3.5.7 Dust. The weapon system shall operate while being subjected to blowing dust with particle size of 0.0001 to 0.0100mm at wind speeds of 15 knots.

3.5.8 Sand. The weapon system shall operate while being subjected to blowing sand of particle size from 0.01 to 1.00mm at wind speed of 35 knots.

3.6 Interchangeability. Unless otherwise specified on the drawings, all parts with the same part numbers shall be interchangeable. Testing shall be as specified in 4.5.3.3.

3.7 Electromagnetic interference characteristics. The weapon system shall meet the applicable electromagnetic interference characteristics requirements of MIL-STD-461 as specified in the contract (see 6.2). Testing shall be as specified in 4.4.

3.7.1 System and radar compatibility. The gun mount (turret) drive electrical noise generated during traversing and stopping the gun mount shall not produce a false radar lock-on. (The ready-to-fire light shall

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remain off). The gun mount (turret) drive electrical noise with the mount traversing drive motors locked and the system in radar mode, the decible (DB) difference while operating before ambient DB, and after revolving hand controls with action switch depressed maximum right and maximum left position, shall not exceed four DB. Distribution box switch shall be in "test" position. The electrical noise shall not produce a false radar lock-on.

3.8 Marking. The weapon system shall be marked in accordance with MIL-W-13855.

3.9 Workmanship. The workmanship shall be in accordance with MIL-W-13855.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. The first article shall be selected from early production and submitted for inspection in accordance with the contract requirements (see 6.2). The first article shall be representative of the production processes to be used during quantity production. The first article shall be subjected to the Quality conformance inspection (see 4.5), Towing test (see 4.6.21 and 4.6.22), Environment test (see 4.6.23), Interchangeability test (see 4.6.24), Electromagnetic Interference test (see 4.7) and such other inspection as necessary to determine that all requirements of the contract have been met.

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4.5 Quality conformance inspection.

4.5.1 Inspection lot. Unless otherwise specified, the formation, size and presentation of inspection lots shall be in accordance with MIL-STD-105. Inspection lots shall be as large as practicable considering manufacturing conditions and suppliers delivery schedule and within the limits of MIL-W-13855. Weapon systems shall be assembled from component parts that have met all inspection requirements specified herein.

4.5.2 Examination.

4.5.2.1 Component parts and concurrent repair parts. Examination of component parts and concurrent repair parts shall be performed in accordance with the criteria specified in the contract (see 6.2). The contractor's examination of these parts shall be accomplished prior to their assembly into the end item or submission for acceptance as repair parts.

4.5.2.2 Weapon systems. Each weapon system shall be visually and manually examined for proper cleaning, presence of protective coatings, completeness of manufacture, freedom of all operating parts through full travel, for security of parts and fasteners, hydraulic oil leakage, rear light assemblies, wheel nut torque, handbrakes and tire pressure, workmanship. Weapon systems failing to meet the requirements specified herein shall be rejected.

4.5.2.2.1 All markings required by final assembly drawings shall be visually examined for clarity and legibility. The identification shall be examined for compliance with the applicable drawing and for secure assembly.

4.5.2.3 Final examination. Final examination of each weapon system shall be performed after completion of all testing and just prior to preservation and packaging. Each weapon system shall be examined as specified in 4.5.2.2. Weapon systems failing the requirements specified herein shall be rejected.

4.5.2.4 Inspection approval stamp. Weapon systems which have passed all examinations after successful completion of testing shall be stamped with the Department of Defense complete inspection approval stamp as specified on the applicable drawing. The Government representative shall control the stamp used for such stampings to assure successful completion of all required examination and test.

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4.5.3 Testing.

4.5.3.1 Failure data. Unless otherwise specified herein, all tests shall be conducted on a complete weapon system exclusive of Government furnished property. If test requirements cited herein are not met, acceptance of the weapon system shall be deferred and the contractor shall accomplish, as applicable, the following action:

a. Conduct a failure analysis study performing a dimensional, physical and visual examination of the components which are suspected to be the cause of failure or malfunction.
b. Evaluate and correct the applicable production processes and procedures to prevent recurrence of the same defect(s) in future production.
c. Examine weapon systems, partially assembled weapon systems and components (including components and subassemblies at in process or final assembly) to insure that material containing the same defect is purged from the inventory and not presented to the Government for acceptance.
d. Submit the results of the failure analysis and the corrective actions taken to the Government for review and approval prior to submitting a reconditioned lot or reconditioned weapon system for retest.

4.5.3.2 Weapon system testing. Unless otherwise specified (see 6.2), each weapon system shall be subjected to the test methods specified in 4.6.1 through 4.6.20, except only one weapon system randomly selected by the Government representative from each 20 systems produced, shall be subjected to the -40°F warm-up temperature (see 3.3.1) using the test method specified in 4.6.1. Failure of any weapon system to meet the requirements of 3.3.1 through 3.3.18 shall be cause for rejection and the contractor shall perform the corrective action as specified in 4.5.3.1.

4.5.3.3 Interchangeability testing. The interchangeability test shall be performed on three weapon systems selected by the Government representative from each 20 systems produced using the test method specified in 4.6.24. After completion of the test and the weapon systems are reassembled, each system shall be subjected to the testing specified in 4.5.3.2. Failure of any reassembled system to meet the requirements (see 3.6) shall be cause for rejection and the contractor shall perform the corrective action specified in 4.5.3.1. Disposition of the interchangeability tested systems shall be as specified in the contract. (see 6.2).

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4.5.4 Packaging examination and testing. Unless otherwise specified (see 6.2), the packaging examination and testing shall be in accordance with MIL-P-14232.

4.5.5 Inspection equipment.

4.5.5.1 Acquistition, maintenance and disposition. Unless otherwise specified (see 6.2), responsibility for acquisition, maintenance and disposition of acceptance inspection and test equipment prescribed on the List of Inspection Equipment Drawing IEL8430372, and for all other inspection equipment required to perform inspection prescribed by applicable specifications shall be in accordance with MIL-I-45607.

4.5.5.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used it must be capable of repetitive measurements to an accuracy of 10% of the total tolerance of the characteristics being inspected.

4.6 Test methods.

4.6.1 Warm-up test. The weapon system shall warm-up to permit operation in the test, radar, manual, external and ground modes to determine compliance with 3.3.1.

4.6.2 Test mode test. The weapon system shall be operated in the test mode. A simulated problem shall be presented to the sight current generator and correctly solved, as indicated by lighting of the GOOD TEST light to determine compliance with 3.3.1.1.

4.6.3 Radar mode test. The weapon system shall be operated in the radar mode (see 6.7). The radar mode and lead computing feature shall allow the gun mount to rotate and track the target. The READY-TO-FIRE indicator lamp, located on the sight, shall light to determine compliance with 3.3.1.2. After release of the trigger switch, action switch and the radar radiate foot switch, the weapon system shall return to the standby condition.

4.6.4 Manual mode test. The weapon system shall be operated in the manual mode. The TARGET and TARGET SPEED controls shall be set at estimated values. The sight gyro shall be electrically caged by depressing the sight cage button during high speed turret slew. When the target is acquired in the sight reticle, the sight cage button shall be released and tracking shall continue until the target is within nominal firing pattern for one second. The READY-TO-FIRE indicator lamp shall remain lit to determine compliance with 3.3.1.3.

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4.6.5 External mode test. The weapon system shall be operated in the external mode. The weapon system shall receive and use range data supplied by the external range control. The READY-TO-FIRE indicator lamp shall light when the external READ-TO-FIRE button is depressed to determine compliance with 3.3.1.4.

4.6.6 Ground mode test. The weapon system shall be operated in the ground mode. The sight shall be mechanically caged seven mils super elevation and the reticle lamp of the sight shall light to determine compliance with 3.3.1.5.

4.6.7 System reaction time test. When subjected to the conditions of 3.3.2, the time required for the weapon system to slew 180° acquire, track and simulate fire on the target shall be measured to determine compliance with 3.3.2.

4.6.8 System fire control test. The system fire control operation test shall be as specified in MIL-F-60664 (see 3.3.3).

4.6.9 Firing test. The firing rate switch on the control panel shall be rotated to the high and low rate and the weapon system fired in each position to determine compliance with 3.3.4 and 3.3.5. At the high rate, fire the four burst limits to determine compliance with 3.3.6.

4.6.10 Field of fire test. The mount and gun shall be operated through all ranges of elevation and azimuth to determine compliance with 3.3.7. The elevation electrical limit switches shall prevent the weapon system from being driven into the mechanical stops.

4.6.11 Slewing rate and acceleration test. The mount and gun shall be slewed in all operational modes. The slew rate and acceleration shall be measured to determine compliance with 3.3.8 and 3.3.9.

4.6.12 Stall and stop test. The weapon system shall be operated through ranges of evaluation and azimuth rotation (see 3.3.7 and 3.3.10). The gun mount shall be driven into the elevation stops at maximum slew rate. The mount and gun shall incur no damage and shall not be capable to slew with the gun in positive travel lock position to determine compliance with 3.3.10.

4.6.13 Firing interrupter test. The firing interrupter assembly (see 3.3.11) shall be visually and manually inspected for secure and correct assembly in the system. Using Government approved test equipment and by simulated firing of the system, automatic interruption of firing shall be checked for each 20° segment of azimuth and each 5° increment of elevation within each segment of azimuth. To assure maxi-

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mum safety for simulated firing, set controls for interrupted firing and ground mode. Set rods for alternately firing in 20° segments and rotate turret through azimuth for each 5° increments of elevation. Observe that the firing voltage test lamp is "off" or "on" during alternate 20° segments corresponding to the rod settings in the firing interrupter assembly. Failure of the firing interrupter to follow the rod settings shall be cause for readjustment of the interrupter assembly and retest.

4.6.14 Azimuth indicator test. The azimuth indicator assembly (see 3.3.12) shall be visually and manually inspected for secure and correct mounting, for free rotation of the dial from 0 through 64 in a clockwise direction and free rotation in a counter-clockwise direction. Using system power and hand controls, position the turret to a definite reference point and set azimuth indicator to 64 on the dial. Slew turret and position at 90° from reference. Continue rotation and reposition on reference point. Compare turret position and indicator readings for accuracy of one half sector on the dial. Repeat the test by rotating turret in opposite direction.

4.6.15 Operational capability test. The weapon system shall be operated as described in 3.3.1, 3.3.3 and 3.3.4 to determine compliance with 3.3.13.

4.6.16 Declutching feeder test. The weapon system shall be fired and ammunition fed to the gun to determine compliance with 3.3.14.

4.6.17 Clearing test.

4.6.17.1 Gun clear switch test. The weapon system shall be fired. The gun shall automatically clear when the GUN CLEAR switch on the control assembly is placed in the AUTO position. The GUN CLEAR switch placed in the OPERATOR position, the gun shall clear by pressing the action switch and actuating the solenoid to determine compliance with 3.3.15.1.

4.6.17.2 Brake-clear and brake switch test. The weapon system shall be checked for proper operation with the brake-clear and brake switch in each position to determine compliance with 3.3.15.2. Dummy cartridges shall be used to check for proper gun and declutching feeder operation while the power system is on. Placing the switch in BRAKE position will allow the gun drive motor to release and dummy rounds to be cycled through the gun by manually rotating the barrel cluster. Placing the switch in the BRAKE-CLEAR position will allow the gun drive motor brake to release and energize the declutching feeder, preventing the rounds from being fed through the gun. When the switch is in center OFF position the brake will prevent the gun from being manually rotated.

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4.6.18 Sustained use test. The weapon system shall be tested for degradation in excess of that specified or other indications of excessive power loss in the weapon system which show noncompliance with 3.3.16 and shall be cause for rejection.

4.6.19 Boresighting. The weapon system shall be subjected to the following boresighting procedure to determine compliance with 3.3.17.

1. Check that the GUN POWER switch is OFF.
2. Set the SYSTEM POWER switch to the ON position and set MODE switch to the RADAR position.
3. Check that the gun is clear. To clear gun, position the BRAKE-CLEAR AND BRAKE switch in the CLEAR AND BRAKE position and rotate the barrels two complete revolutions. Remove any rounds in the case chute.
4. Time the gun in the No. 1 firing position. Hold the BRAKE-CLEAR AND BRAKE switch in the CLEAR AND BRAKE position, depress the gun index pin, and locate the No. 1 barrel (marked on the muzzle clamp) in the firing position which is at the 8 o'clock position when viewed from the breech end. Release the BRAKE-CLEAR AND BRAKE switch and index pin. Check that the index pin has returned to normal.
5. Unstow the radar antenna, if stowed, by loosening the securing bolt and rotating antenna arm assembly counter-clockwise as far as possible. Re-install securing bolt.
6. Check to see that the MODE switch is set to NORMAL and the MAINT switch is ON.
7. Select a distant aiming point (at least 2500 meters away) and depress an action switch and train the gun to the approximate position for boresighting.
8. Set the switch on the distribution box to STATIC.
9. Insert the muzzle adapter from the boresight kit into the muzzle of the No. 1 barrel.
10. Install the boresight telescope in the mandrel assembly and secure with socket head screws on the mandrel assembly.
11. Insert the mandrel assembly into the muzzle adapter with the eyepiece of the telescope in the horizontal position and press to tighten the adapter and mandrel assembly in the barrel.
12. Release the two azimuth drive motor brakes and the elevation drive motor brake by positioning the brake release levers perpendicular to the motor axis.
13. Sight through the boresight telescope on a distant point. Rotate the mandrel assembly and boresight telescope 90° to the horizontal position on the opposite side of the barrel.

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Sight through the boresight telescope to determine that the point sighted on initially is still at the crossed hairs. If the cross hairs are displaced over 5 mils from the point sighted on, the boresight kit is defective, improperly assembled, or improperly mounted. Compensate for under 5 mil error by offsetting by one half of the error from the boresight target.

14. Manually position the gun so the cross hairs of the boresight telescope are 3.5 mils to the right and 3 mils below the distant aiming point.
15. Lock the azimuth and elevation drive motor brakes and check that the gun is still properly aimed.
16. Manually uncage the XM61 sight.
17. Adjust the LAMP control assembly for a clear reticle in the sight.
18. Remove only the front and right caps on top of the sight.
19. Adjust the socket head bolts on the top of the sight to center the inner reticle of the sight on the distant aiming point. The front adjustment is for elevation and the right adjustment is for azimuth. Move the eye up, down, left and right to check for parallax (Parrallax is displacement between reticle and range.) If excessive parallax is experienced, refer to the sight organizational maintenance.
    NOTE: A second crew member must check through the boresight telescope while the adjustments are being made to assure that the gun does not move.
20. Install the caps on the sight.
21. Remove the boresight telescope, mandrel assembly, and install the antenna boresight telescope in the mount provided in the antenna. Remove muzzle adapter from No. 1 barrel.
22. Loosen the clamp and pull back the protective cover on the inboard end of the gun elevation input shaft coupling to the radar antenna. Loosen the set screws holding the universal joint to the shaft. Rotate the radar input shaft to move the horizontal cross hair of the telescope reticle on the distant aiming point. Tighten the universal joint set screws to 75 to 85 inch-pounds torque making sure that the joint is fully inboard. Reposition the protective cover and tighten the clamp.
    NOTE: A second crew member must check that the sight remains centered on the distant aiming point.
23. Remove the access cover from the traverse boresight adjustment which is located on the reflector and feed housing assembly. Loosen the lock and turn the adjustment screw until the verticle cross hair on the telescope reticle intersects the distant aiming point, then tighten the lock.

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NOTE: A second crew member must check that the sight remains centered on the distant aiming point.

24. Repeat steps 22 and 23 if necessary to center the reticle cross hairs.
25. At the antenna stow control, place the MODE toggle switch to STOW.
26. Using the TRAV toggle switch, drive the antenna to the left approximately 10 degrees.
27. Place the MODE toggle switch to NORMAL.
28. At the boresight telescope mounted on the antenna, observe that the cross hairs within the reticle are positioned on the distant aiming point.
29. Repeat steps 25 through 28, except in step 26, drive the antenna to the right.
30. At the antenna stow control, place the MODE toggle switch to STOW.
31. Using the ELEV toggle switch, drive the antenna up approximately 10 degrees.
32. Place the MODE toggle switch to NORMAL.
33. At the boresight telescope mounted on the antenna, observe that the cross hairs within the reticle are positioned on the distant aiming point.
34. Repeat steps 30 through 33, except in step 33, drive antenna down approximately 10 degrees.
    NOTE: If, when performing step 27 and step 32 the antenna does not reposition automatically on the distant aiming point (ref 28 and 29), a malfunction exists.
35. Replace the access cover on the traverse boresight adjustment and the gun elevation input shaft protective cover.
36. On the antenna stow control, set the MAINT switch to OFF.
37. Remove the boresight telescope from the antenna reflector.
38. Cage the sight.
39. Set the SYSTEM POWER switch to the OFF position.
40. Set the switch on the system distribution box to NORMAL.
41. Recheck that there is no boresight equipment in the muzzle.

4.6.20 Hand brake test. The weapon system shall be subjected to the hand brake test to determine compliance with 3.3.18, unless, documentary evidence is supplied as proof that the brakes were tested during production by an equivalent static torque test with the brake assembly mounted to the running gear.

4.6.21 Towing test. The weapon system shall be towed by the XM705, XM715 or M561 2½ ton truck, when coupler adapter (8438335) is installed on the carriage, over terrain and road conditions at speeds specified in Table I to be in compliance with 3.4.

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4.6.22 Fording test. The weapon system shall be submerged to a 30 inch fording depth with water leakage into the carriage not exceeding 473 cubic centimeters per hour to be in compliance with 3.3.21.

4.6.23 Environmental test. The weapon system shall be tested as below to determine compliance with 3.4.1 through 3.4.9. Where advantageous, naturally occurring environments may be used in lieu of laboratory produced conditions. There shall be no evidence of corrosion, degradation of performance other than as specified, or damage that would prevent the weapon system from performing its intended function.

4.6.23.1 Altitude test. The weapon system shall be tested for altitude to determine air transportability to 40,000 feet and operability to 10,000 feet in accordance with Procedure I, Method 500 of MIL-STD-810.

4.6.23.2 High temperature test. The weapon system shall be tested when stabilized at 125°F after four hours in a storage temperature at 155°F in accordance with Procedure I, Method 501 of MIL-STD-810. The weapon system shall be tested for operation at 120°F in accordance with Procedure II, Method 501 of MIL-STD-810.

4.6.23.3 Low temperature test. The weapon system shall be tested for a low temperature storage condition of -40°F for 12 hours in accordance with Procedure I, Method 502 of MIL-STD-810. At 0°, -20°, and -40°F, the degradation in performance shall be not more than 5, 10, and 15 percent, respectively.

4.6.23.4 Rain test. The weapon system shall operate in rain and wind conditions in accordance with Procedure I, Method 506 of MIL-STD-810,

4.6.23.5 Humidity test. The weapon system shall be subjected to humidity conditions for seven cycles of 12 hours each in accordance with Procedure I, Method 507 of MIL-STD-810. The weapon system shall withstand storage at 100 percent relative humidity at 85°F for 48 hours and shall operate at this relative humidity at temperature of 85°F.

4.6.23.6 Fungus test. The weapon system shall operate after being subjected to the fungus test in accordance with Procedure I, Method 508 of MIL-STD-810.

4.6.23.7 Salt fog test. The weapon system shall operate after being subjected to the salt fog test in accordance with Procedure I, Method 509 of MIL-STD-810.

4.6.23.8 Dust test. The weapon system shall operate when subjected to blowing dust, with particle size from 0.0001 to 0.01 mm and wind speeds of 15 knots in accordance with Procedure I, Method 510 of MIL-STD-810.

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4.6.23.9 Sand test. The weapon system shall be operated at speeds up to 30 miles per hour in sandy terrain with wind speeds of 35 knots. The weapon system shall be operated, with the vehicle both stationary and moving, in blowing sand having a particle size from 0.01 to 1.00mm and with wind speeds of 15 knots or greater. Laboratory produced conditions shall be permitted for this test.

4.6.24 Interchangeability test. Interchangeability test procedures shall provide for three weapons systems to be disassembled and reassembled in a prearranged pattern from the parts of each weapon system, divided into three groups of non-mating parts and distributed in three different areas, each containing a complete weapon system. The exchange of parts and assemblies shall result in the maximum exchange in mating parts. The interchangeability test shall include major subassemblies and repair parts for major subassemblies as specified in Table III.

TABLE III. INTERCHANGEABLE ITEMS

EQUIPMENT	PART NUMBER
Support sight assembly	8436972
Support assembly, equilibrator (2)	8438318
Brake and clear switch assembly	8437323
Drive assembly, elevation	8436847
Drive assembly, azimuth	8437053
Azimuth indicator assembly	8438753
Recoil adapter assembly (2)	8436802
Muzzle clamp assembly	8436804
Declutching feeder assembly	8438645
Servo amplifier assembly (Az. Master)	8436842
Distribution box assembly	8438325
Sight current generator	8438320
Slip ring assembly	8438355
Running gear assembly	8438691

4.7 Electromagnetic interference characteristics test. The weapon system shall be tested in accordance with the test methods of MIL-STD-462 to determine compliance with the requirements of 3.7.

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4.7.1 System and radar compatibility test. With the system in the "RADAR MODE" the gun mount shall be traversed at maximum slew rate in both clockwise and counterclockwise direction. (The ready to fire light shall remain off to determine compliance with 3.7.1.) The gun mount electrical noise (radiated emission) shall be measured in an ambient condition, and during traversing the gun mount. The radiated emission measurement shall be taken in accordance with MIL-STD-462, Method REO5 at the broad band frequencies of 150KHz to 1 GHz. The decible difference shall not exceed four (4) DB in accordance with the requirements of 3.7.1.

5. PREPARATION FOR DELIVERY

5.1 Pilot pack. A pilot pack shall consist of a complete gun mount packaged in accordance with the Packaging Data Sheet P8430372 for the level of protection specified in the contract (see 6.2), packed level C and forwarded in accordance with 3.1.

5.2 Preservation, packaging, packing and marking. Guns shall be preserved, packaged, packed and marked in accordance with the Packaging Data Sheet P8430372 for the level of protection specified in the contract (see 6.2).

5.3 Repair parts. Repair parts shall be preserved, packaged, packed and marked in accordance with the Packaging Data Sheet or the requirements and for the level of protection specified in the contract (see 6.2).

6. NOTES

6.1 Intended use. The M167 Towed Gun, 20mm, Antiaircraft Artillery is intended for, but not limited to close range antiaircraft cover or strafing operation.

6.2 Ordering data. Procurement documents should specify:

6.2.1 Procurement data.

a. Title, number and date of this specification.
b. List of drawings and specifications pertinent to the weapon subsystem showing applicable revision dates.
c. Applicable electromagnetic interference characteristics requirements (see 3.7 and 4.7).

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d. Inspection criteria for components (see 4.5.2.1).
e. Weapon system testing, if different (see 4.5.3.2).
f. Packaging examination and testing, if different (see 4.5.4).
g. Responsibility for acquisition, maintenance, and disposition of inspection equipment, if other than as specified (see 4.5.5.1).
h. Fire control criteria desired (see 3.3.3).
i. Disposition instructions for interchangeability tested weapon system, parts and rejected material (see 4.5.3.3).
j. Shipping instructions for first article and pilot pack (see 3.1 and 5.1).
k. Selection of applicable levels of preservation, packaging and packing (see 5.1 and 5.2).
l. Packaging instructions for repair parts (see 5.3).

6.2.2 Contract data requirements. Requirements for monthly reports of the results of final examination, performance characteristics and interchangeability tests shall be specified on a DD Form 1423 included in the contract.

6.2.3 Management control requirements. When warranted, the contract should specify the application of MIL-I-45208 or MIL-I-9858 on the Management Control Systems Summary List, DD Form 1660.

6.3 Serial numbers. The contractor should be furnished a block of serial numbers by the contracting officer to be used for marking the weapon system identification plate.

6.4 Government furnished equipment. Government furnished equipment shall be as specified in the contract (see 6.2).

a. Range radar set, AN/VPS-2, part number 10548619.
b. Generator set, gasoline engine, part number 8438257.
c. Lead computing sight, M61, part number 10545010.
d. Night vision sight, crew served weapons, AN/TVS-2B.
e. Sight telescope, M134, part number 11728190.
f. Telescope mount, M164, part number 11728180.
g. Gunner's quadrant, M1A1.
h. Muzzle clamp assembly, part number 11010336, FSN 1005-754-5291.

6.5 Supersession data. This specification includes the requirements of U.S. Army Weapons Command Purchase Description, USAWECOM PD-22, dated 30 September 1970.

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6.6 Unless otherwise specified (see 6.1 h), the contract should specify the application of MIL-I-45607 and MIL-C-45662 on the Management Control Summary List, DD Form 1660.

6.7 Radar caution: Caution shall be exercised during operation of the radar system due to a radiation hazard to personnel (see 4.6.3).

Custodian:
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Preparing activity:
Army - WC

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